A Theory of Meaning, Working Memory, and the Structure of Conscious Experience

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Abstract

We propose a theoretical framework in which meaning, working memory, and conscious experience arise from wave dynamics on a weighted graph representing the structural connectome. Three principal contributions are made. First, we identify working memory with a time-varying perturbation $\delta L(t)$ of the graph Laplacian, governed by Hebbian co-activation: $d(\delta L)/dt = -\eta C(\psi) - \kappa \delta L$, where $C(\psi)$ is the sparse co-activation matrix of the current wave state ψ. This yields a mechanistic account of working memory capacity (Miller's magical number 7), the phonological loop, sleep-based consolidation, and the Default Mode Network — all as emergent properties of field-Laplacian coupling. Second, we establish a categorical distinction between weight change and topological change in L: weight modification shifts eigenmodes continuously (deepening known attractors), while edge addition creates new paths in the graph enabling interference patterns that were geometrically impossible before, and thus qualitatively new attractors. We identify this distinction with the difference between learning and understanding. Third, we propose that conscious experience is the rate of change of the global wave state: $E \propto |d\psi/dt|$. This formulation — experience as the felt change in the wave state, not the state itself — yields mechanistic accounts of habituation, the orienting response, emotional intensity, meditation, and priming, and maps structurally onto Husserl's phenomenological time-structure (retention, primal impression, protention). We further show that the binding problem dissolves under Axiom 1 (no node carries meaning; meaning is a global field state), and that the reasoning loop — cyclic re-injection of Transformer output into the field — instantiates Baddeley's phonological loop as a physical self-sustaining cycle. Detailed implementation parameters will be reported in a companion technical paper; a prototype implementation is feasible on a single modern GPU.

Keywords: graph neural fields, connectome harmonics, working memory, Hebbian learning, consciousness, wave dynamics, graph Fourier transform, semantic emergence

1. Introduction

How does the brain give rise to meaning, memory, and experience? These three questions have been addressed separately by decades of research, yet a unified mechanistic framework connecting all three remains absent. We argue that such a framework emerges naturally from treating the connectome as a weighted graph and analysing wave dynamics on that graph via the Graph Fourier Transform (GFT).

The key insight is that meaning is not stored — it resonates. Incoming signals interact with the learned topology of the connectome (the structural graph), and meaning emerges as the attractor state of the

resulting wave dynamics. This shift from representational to resonance-based semantics dissolves several classical problems, including the binding problem, and provides mechanistic accounts of phenomena that have resisted explanation: the capacity limit of working memory, the Default Mode Network, why sleep consolidates long-term memory, and the phenomenal structure of conscious experience.

The framework rests on three axioms and three principal contributions:

1. Axiom 1 (Global Meaning): No node carries meaning. Meaning is a global wave state $\psi(t)$ over all n nodes simultaneously. This is empirically motivated by Lashley's finding that memory survives destruction of large cortical areas [18].
2. Axiom 2 (Topology as Memory): The connectome does not store meanings; it defines a topology determining which global wave patterns can stably resonate.
3. Axiom 3 (Spectral Readout): The FieldReader is a spectral projection $\hat{\psi} = \Phi^T\psi$, not spatial downsampling.

The three principal theoretical contributions are: (i) a field-Laplacian coupling equation that integrates working memory into the wave dynamics; (ii) a spectral characterisation of the learning–understanding distinction; and (iii) the identification of conscious experience with $|d\psi/dt|$.

2. Background and Related Work

2.1 Graph Neural Fields

Aqil et al. [1] established that wave dynamics on the human connectome graph, implemented via the graph Laplacian, reproduce the empirically observed harmonic power spectrum of resting-state fMRI data. The eigenvectors of the connectome Laplacian — termed connectome harmonics — correspond to functionally relevant resting-state networks. Abdelnour et al. [2] further demonstrated that spectral graph theory predicts alpha- and beta-band oscillation patterns from structural connectivity alone. Our framework builds directly on this substrate, extending it with Hebbian learning [13], coupled Laplacian dynamics, and a cognitive interpretation of the eigenstructure.

2.2 Memory Systems

The classical three-store model of Atkinson and Shiffrin [3] distinguishes sensory register, working memory, and long-term memory. Baddeley's working memory model [4, 17] proposes a phonological loop, visuospatial sketchpad, and central executive. Miller [5] established the capacity limit of approximately 7 ± 2 items. McClelland et al. [6] identified the hippocampus with rapid encoding and the neocortex with gradual extraction. None of these accounts provide a single mechanistic equation. Our framework derives all three memory systems from the coupled dynamics of ψ and L.

2.3 Consciousness Research

Tononi [7] proposed integrated information theory (Φ), identifying consciousness with integrated information. Dehaene [8] developed the global workspace theory, in which consciousness corresponds to global availability of information. Chalmers [9] formulated the hard problem: why do physical processes give rise to subjective experience at all? Our proposal — $E \propto |d\psi/dt|$ — does not solve the hard problem but provides a precise functional formulation that connects to Husserl's phenomenological structure [10] and yields testable predictions.

3. Theoretical Framework

3.1 The Connectome as a Weighted Graph

Let $G = (V, E)$ be an undirected weighted graph with $n = |V|$ nodes and sparse edge set E. The weighted adjacency matrix is $A \in \mathbb{R}^{n\times n}$, the degree matrix $D = \mathrm{diag}(\sum_j A_{ij})$, and the graph Laplacian $L = D - A$. Because L is real symmetric and positive semi-definite, its eigenvectors $\Phi = [\varphi_1, \ldots, \varphi_n]$ form an orthonormal basis:

$$L\varphi_k = \lambda_k\varphi_k, \quad 0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$$

The Graph Fourier Transform (GFT) of a graph signal $\psi \in \mathbb{C}^n$ is $\hat{\psi} = \Phi^T\psi$, with inverse $\psi = \Phi\hat{\psi}$ [15]. Low eigenvalues correspond to globally smooth patterns (semantic structure); high eigenvalues correspond to locally varying patterns (noise and fine detail). The FieldReader truncates to the K lowest modes: context = $\hat{\psi}[:K]$, providing a K-dimensional spectral summary for a Transformer readout.

3.2 Wave Dynamics: The Complete Equation

The wave state $\psi(t) \in \mathbb{C}^n$ evolves on the connectome graph across five frequency bands (corresponding biologically to the delta, theta, alpha, beta, and gamma rhythms), each with a characteristic time constant. Separating the Laplacian into a slowly evolving base L_0 (long-term memory) and a rapidly adapting perturbation $\delta L(t)$ (working memory), the complete coupled system is:

$$d\psi/dt = -i(L_0 + \delta L)\psi - \gamma\psi + S(t) \qquad (1)$$

$$d(\delta L)/dt = -\eta C(\psi) - \kappa\delta L \qquad (2)$$

where γ is the damping constant, $S(t)$ is the injected signal (encoder output), η is the Hebbian learning rate, κ is the forgetting rate, and $C(\psi)$ is the sparse co-activation matrix with $C_{ij} = |\psi_i||\psi_j|$ for $(i,j) \in E$ and 0 otherwise. The single free parameter governing working memory strength is $\alpha = \eta/\kappa$. This system contains all three classical memory stores: $\psi(t)$ is the sensory register (milliseconds); $\delta L(t)$ is working memory (minutes, reversible); L_0 is long-term memory (hours to years, structural).

3.3 Numerical Integration

The nonlinear coupling between ψ and δL requires a time-consistent integrator. We employ a second-order Magnus integrator [16] with self-consistent midpoint evaluation (two Newton iterations), reducing the local truncation error to $O(\Delta t^3)$ compared to $O(\Delta t)$ for a naive sequential update. Wave propagation is computed via the Chebyshev polynomial approximation of the matrix exponential $\exp(-iL_{eff}\Delta t)$ [15], with cost $O(K_{cheb} \times nnz)$ per time step — linear in graph edges, independent of n. The equation for δL uses an Euler step, which is accurate because δL evolves on a much slower timescale than ψ.

3.4 Signal Injection and the Reasoning Loop

An input signal is encoded to a vector v, projected into the spectral basis via the K lowest eigenvectors of L_0, and injected as a global initial wave pattern. This satisfies Axiom 1: injection creates a pattern over all n nodes, not a localised activation.

The reasoning loop closes the system: the readout Transformer's output (natural language) is re-encoded and re-injected into the field. Each generated thought becomes the next input signal. This self-sustaining cycle implements inner speech as a physical dynamical process: the Transformer generates, the encoder re-encodes, the field accumulates intermediate results as wave interference — not as an explicit list. This is Chain-of-Thought as genuine temporal dynamics, distinct from token-sequence pseudo-time in standard language models.

4. Three Principal Contributions

4.1 Contribution I: A Mechanistic Account of Working Memory and Related Phenomena

The field-Laplacian coupling equations (1)–(2) provide a unified mechanistic account of several classical phenomena:

Miller's capacity limit. Multiple attractors compete for field energy. When the total co-activation energy of simultaneously maintained attractors exceeds the available field energy, all attractors collapse. The number 7 ± 2 is not a parameter but emerges from the balance between attractor depth and field dissipation. The observation that experts maintain more domain-specific items (deeper attractors, less energy per item) follows directly.

The phonological loop. Baddeley's loop [4, 17] is the reasoning loop operating on a fixed attractor: the Transformer generates the item, the encoder re-injects it, the field is refreshed. Item loss occurs when the loop latency exceeds the attractor decay time $1/\kappa$. This is not a metaphor — it is equation (1) with a specific input signal $S(t)$ equal to the encoded current item.

Sleep consolidation. During deep sleep ($S(t) = 0$), equation (2) drives δL toward zero while strong co-activations accumulated during waking push permanent changes into L_0 via a long-term Hebbian rule [13]. Strong edges are stabilised; weak edges are pruned. The transition from working memory to long-term memory is a topological phase transition, not a copying operation.

The Default Mode Network. At rest ($S(t) = 0$), the coupled system (1)–(2) has a fixed point determined by the learned L_0. The spontaneous activity pattern at this fixed point corresponds to the DMN: the system oscillates on what it knows. Individual differences in DMN topology reflect individual differences in L_0, predicting that the DMN should be altered by any intervention that modifies L_0, including sustained meditative practice and neurodegeneration.

Priming. Semantically related concepts have overlapping eigenmode decompositions. The residual activation of a priming stimulus in $\psi(t)$ facilitates attractor formation for the target concept, reducing formation time. For active priming maintained through the reasoning loop, this facilitation persists for seconds, not just milliseconds — explaining why priming effects exceed the timescale of simple neural decay.

4.2 Contribution II: Learning versus Understanding as Eigenstructure Change

We establish the following distinction via graph spectral theory:

Weight modification ($dA_{ij} \neq 0$ for an existing edge): shifts all eigenvectors continuously. Existing attractors deepen or shallow. No qualitatively new attractor becomes accessible. This corresponds to learning by repetition: the same patterns become more reliable.

Edge addition ($A_{ij}: 0 \rightarrow w$ for a new edge): creates a new path in the graph, enabling interference patterns that were geometrically impossible before. Qualitatively new attractors can form. This corresponds to understanding: a genuinely new connection enables genuinely new thought.

This distinction is categorical, not gradational. The subjective phenomenon of the Aha moment corresponds to the first stable occupation of a new point in eigenmode space — not the creation of a new eigenmode, but a new stable superposition becoming accessible under the expanded topology. Long-term consolidation (edge addition during sleep) makes this accessible superposition permanent. Weight changes occur on the timescale of η/κ (minutes); edge addition requires repeated co-activation over many episodes, followed by sleep consolidation.

We note that whether semantic structure genuinely emerges from a homogeneous initial connectome under Hebbian learning alone is an empirical question addressed by simulation, not assumed by the theory. The theory predicts emergence is possible; whether it occurs must be verified experimentally.

4.3 Contribution III: Experience as the Rate of Change of the Wave State

We propose:

$$E(t) \propto |d\psi/dt| \qquad (3)$$

that is, the intensity of conscious experience at time t is proportional to the magnitude of the time derivative of the global wave state. This formulation makes the following predictions:

Habituation. When a stimulus produces a stable attractor, $d\psi/dt \to 0$. Experience of that stimulus fades while the system continues to process it. This matches the neural signature of habituation and the everyday observation that constant stimuli cease to be noticed.

Orienting response. A novel stimulus disrupts the current attractor, producing large $|d\psi/dt|$, matching the P300 component in EEG.

Emotional intensity. Rapid attractor transitions produce very large $|d\psi/dt|$. Emotional intensity is a property of the speed of field change, not the content of experience — consistent with the observation that anticipation of an event is often more intense than the event itself.

Meditation and flow. Deliberate stabilisation of the wave state reduces $|d\psi/dt|$ toward a quiet minimum. The flow state corresponds to uniform field development: optimal experience between boredom ($|d\psi/dt| \approx 0$) and overwhelm ($|d\psi/dt|$ very large).

Phenomenological correspondence. Husserl's time-structure of consciousness — retention, primal impression, protention [10] — maps directly: retention = $\psi(t - dt)$, primal impression = $d\psi/dt$, protention = the direction of the current attractor. This is structural identity, not analogy.

We emphasise that equation (3) does not solve the hard problem [9]. It does not explain why $d\psi/dt$ should feel like anything. It replaces the question 'why is there experience?' with the sharper question 'why does field change feel like something?' and generates testable predictions that existing theories do not.

5. The Binding Problem Dissolved

The binding problem [11] asks how separately processed features (colour, shape, motion) are unified into a single percept. The classical answer — gamma synchrony — explains when binding occurs but not why synchrony should produce unity.

Under Axiom 1, the binding problem is dissolved rather than solved. A 'red moving ball' is not a composition of RED + ROUND + MOVING. It is a single global wave state ψ whose eigenmode decomposition simultaneously encodes all three aspects. There is nothing to bind because the features were never separate: meaning is global by definition. The binding problem arose from the incorrect premise that features are locally coded and must be re-assembled.

The unity of experience is the unity of $d\psi/dt$: at any moment, there is one global rate of change, one stream of experience, even if it has rich internal structure. This provides a mechanistic basis for the 'subjective unity of perception' without requiring an additional binding mechanism.

6. Empirically Testable Predictions

The framework makes the following predictions, each addressable with existing experimental or computational methods:

P1 (Attractor formation speed). In a simulated system trained on a curriculum of concepts, attractor formation time should decrease monotonically with training episodes for known concepts and remain elevated for novel concepts. The ratio (known / novel attractor formation time) provides a direct measure of semantic competence.

P2 (Working memory capacity). The maximum number of simultaneously stable attractors should scale with field energy divided by average attractor depth. Experts should maintain more domain-specific items, predicting the domain-specificity of working memory capacity beyond the classical universal limit.

P3 (Sleep consolidation). In a system with a simulated sleep phase ($S(t) = 0$), post-sleep attractor formation should be faster and more stable for items learned before sleep than for items without consolidation.

P4 (Experience intensity). Subjective intensity ratings from human subjects performing analogous cognitive tasks should correlate with $|d\psi/dt|$ measured in the model. Novel stimuli should produce higher $|d\psi/dt|$ than familiar ones; emotional stimuli higher than neutral; meditation sessions lower than resting state.

P5 (Topological signature of understanding). The eigenspectrum of the simulated connectome should show discontinuous shifts following insight-type learning events (first successful generalisation to a novel combination) and only continuous shifts following repetition learning.

P6 (Priming timescale). Semantic priming facilitation should persist for seconds when the prime is maintained via the reasoning loop (active priming), and decay within hundreds of milliseconds when the prime is not re-injected (passive priming). This distinguishes $d\psi/dt$ residuals from active loop maintenance.

7. Towards Implementation

The complete system is implementable on a single modern GPU. The graph is initialised as a hierarchical small-world structure [14] biologically motivated by cortical column and cortical area organisation, with sparse long-range connections analogous to association fibres. Wave propagation uses Chebyshev polynomial approximation of the matrix exponential [15], with cost linear in graph edges and independent of n. Eigenvectors are computed periodically using LOBPCG or the Nystrom approximation. With 5 frequency bands and the Magnus integrator [16], attractor formation completes well within one second at prototype scale.

Detailed implementation parameters, graph construction, hardware benchmarks, and training protocol will be reported in a companion technical paper. The theoretical framework is independent of any specific parameter choice; the present paper establishes the conceptual and mathematical foundations.

8. Discussion

The framework presented here differs from existing graph neural field models [1, 2] in three fundamental respects: (i) the Laplacian is not fixed but coupled to the wave state via Hebbian dynamics [13], giving rise to memory as field-topology co-evolution; (ii) meaning is defined as a global wave state rather than a local activation pattern; and (iii) the framework makes commitments about conscious experience that are absent from existing computational models.

The distinction between weight change and edge addition as the mechanistic basis for the learning–understanding distinction is, to our knowledge, novel in the context of graph spectral theory. It predicts a testable discontinuity in the eigenspectrum of the learned connectome that could be observed in simulations and, eventually, in vivo via longitudinal connectome imaging before and after insight-type learning events.

The identification $E \propto |d\psi/dt|$ is consistent with resonance theories of consciousness [12] and provides an operationally precise version of the claim that consciousness is associated with large-scale shared resonance. It differs from Integrated Information Theory [7] in requiring no computation of integration; it differs from Global Workspace Theory [8] in positing no discrete broadcast event. It grounds

phenomenal structure in the continuous dynamics of a physical field and connects to Husserl's phenomenology [10] without invoking non-physical concepts.

Several important questions remain open. Whether semantic structure genuinely emerges from a homogeneous initial connectome under Hebbian learning alone is an empirical question that can only be addressed by simulation. The optimal working memory coupling strength α is not derivable from first principles. The relationship between the model's $|d\psi/dt|$ and human subjective intensity reports requires experimental investigation. The extension to directed graphs remains technically challenging, as the non-symmetric Laplacian loses the clean spectral properties on which the current framework depends.

9. Conclusion

We have presented a theoretical framework unifying meaning, working memory, and conscious experience within wave dynamics on the structural connectome on the structural connectome. The central coupled equations — $d\psi/dt = -i(L_0 + \delta L)\psi - \gamma\psi$, $d(\delta L)/dt = -\eta C(\psi) - \kappa\delta L$ — contain all three classical memory systems as emergent properties of field-Laplacian coupling. The distinction between weight change and edge addition provides a mechanistic account of learning versus understanding that is categorical, not gradational. The identification $E \propto |d\psi/dt|$ connects the framework to phenomenology, resolves the binding problem under Axiom 1, and generates six falsifiable predictions.

The framework makes specific, falsifiable commitments. If attractor formation speed fails to decrease monotonically with learning, or if the eigenspectrum shows no discontinuous shifts at insight-type events, the framework is falsified at the computational level. If $|d\psi/dt|$ fails to correlate with subjective intensity reports in analogous tasks, the phenomenological proposal is falsified. This is its essential scientific character: not that it must be right, but that it can be shown to be wrong in precise, measurable ways.

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